4. Access Control

Computer Security Courses @ POLIMI

What is Access Control?

- A binary decision:
  - Access either *allowed* or *denied*
  - What could *possibly* go wrong?
- Scale goes wrong!
  - You cannot explicitly *list* the answers
  - Need to condense them in *rules*
- Questions
  - How do we *design* the access rules?
  - How do we *express* the access rules in practice?
  - How do we appropriately *apply* them?

Who Does it? The Reference Monitor

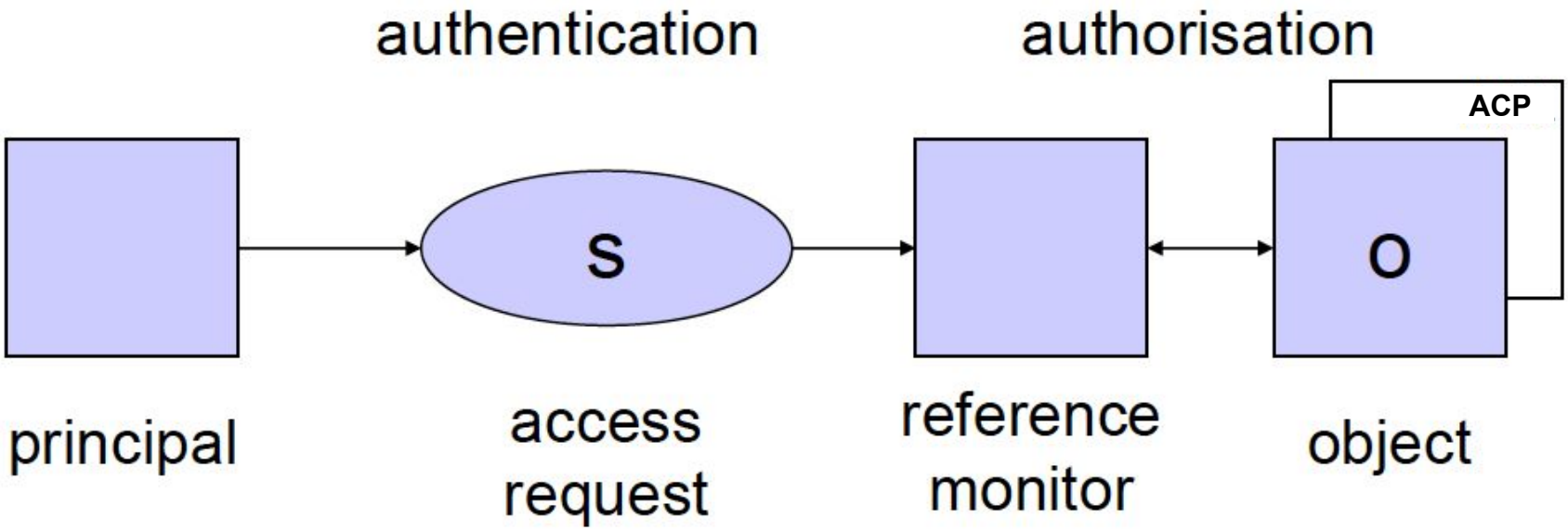
Enforces access control policies ("who does what on which resource"). All modern kernels have a reference monitor implementation.

Requirements for the RM:

- Tamper proof
- Cannot be bypassed
- Small enough to be verified/tested

J. Anderson, 1972 "Computer Security Technology Planning Study"

Authentication & Authorization



Authentication & Authorization

- **Authentication**: reference monitor verifies the identity of the principal (user) making the request:
  - User enters username and password.
  - If the values entered are correct, the user is "authenticated".
    - We could say: "The machine now runs on behalf of the user".
  - This might be intuitive, but it is imprecise.
    - Log on creates a **<u>process</u>** that runs with access rights assigned to the user.
    - Typically, the process runs under the user identity of the user who has logged on.

Authentication & Authorization

- **Authentication**: reference monitor verifies the identity of the principal (user) making the request.
- **Authorization**: reference monitor decides whether access is granted or denied.
  - <u>Reference monitor has to find and evaluate the security policy relevant for the given request.</u>

“Easy” in centralized systems but in distributed systems,

- how to find all relevant policies?
- how to make decisions if policies may be missing?

Users & User Identities

- Requests to reference monitor do not come directly from a user or a user identity, but from a **process**.
- The active entity making a request within the system is called the **subject**.
- You must distinguish between these concepts:
  - **User**: person;
  - **Principal (User identity)**: name used in the system, possibly associated with a user; Active entity.
  - **Subject: Process** running under a given user identity**.**
  - **Object**: Passive entity – file or resource.
  - **Access operations**: Vary from basic memory access (read, write) to method calls in object-oriented systems.

Access control models

Can be roughly divided in

- Discretionary Access Control (DAC)
- Mandatory Access Control (MAC)
- Role-Based Access Control (RBAC)

Difference between DAC and MAC

- who assigns privileges

RBAC abstracts roles from identities

Discretionary Access Control

- Resource <u>owner</u> *discretionarily* decides its access privileges
  - Stefano creates a file
    - Assigns Federico the privilege of reading it
- If this sounds “normal” it is because **all** off-the-shelf OSs implement DAC
  - Windows
  - Linux and other UNIX flavors
  - Mac OS X
  - Also applications, social networks…mostly DAC!

Examples of DAC systems

- UNIX
  - **Subjects**: users, groups
  - **Objects**: files (everything, really)
  - **Actions**: read, write, execute
- Windows (not the 95/98/ME branches)
  - **Subjects:** with *roles* instead of *groups*, multiple ownership of users and roles over files
  - **Objects**: files and “other” resources
  - **Actions**: delete, change permissions, change ownership.

Permissions (r, w, x) **Subjects (user, group)** **Objects (files, dirs, ...)**

```
lrwxr-xr-x    1 phretor  staff      30 Jul  3  2011 .irssi -> /Users/ph
-rw-r--r--    1 phretor  staff     140 Dec  1  2012 .jackdrc
drwxr-xr-x    3 phretor  staff     102 Feb 21  2008 .jmf
-rwxrwxrwx@   1 phretor  staff      41 Dec  9  2012 .khlbshcrc
-rw-------    1 root     staff      51 Oct 27  2008 .lesshst
drwxr-xr-x    4 phretor  staff     136 Jan  4  2008 .lftp
drwx------    4 phretor  staff     136 Jul 22  2009 .links
drwxr-xr-x    3 phretor  staff     102 Apr  1  2013 .m2
lrwxr-xr-x    1 phretor  staff      32 Jul  3  2011 .mailcap -> /Users/
drwxr-xr-x    3 phretor  staff     102 Jan  7  2008 .mldonkey
drwxr-xr-x    4 phretor  staff     136 Jan 25 16:29 .mono
lrwxr-xr-x    1 phretor  staff      31 Jul  3  2011 .mutt.d -> /Users/p
lrwxr-xr-x    1 phretor  staff      36 Jul  3  2011 .muttprintrc -> /Us
lrwxr-xr-x    1 phretor  staff      31 Jul  3  2011 .muttrc -> /Users/p
drwxr-xr-x   11 phretor  staff     374 Jan 31  2008 .ncftp
drwxr-xr-x    8 phretor  staff     272 Dec  7 19:59 .neocomplcache
drwxr-xr-x    8 phretor  staff     272 Oct 21  2012 .neocon
drwxr-xr-x   11 phretor  staff     374 Feb  9  2013 .npm
lrwxr-xr-x    1 phretor  staff      38 Jul  3  2011 .offlineimaprc -> /
drwxr-xr-x   15 phretor  staff     510 Feb  4 22:23 .oh-my-zsh
drwxr-xr-x    6 phretor  staff     204 Apr 20  2013 .parentseye
drwxrwxr-x    3 phretor  staff     102 Dec 24  2010 .pip
drwx------    6 phretor  staff     204 Apr  5  2008 .psi
-rw-------    1 phretor  staff      90 Mar 17  2010 .psql_history
```

UNIX Permissions

Mode | Owner | Group | File Size | Last Modified | Filename

```
drwxrwxrwx 2 sammy sammy       4096 Nov 10 12:15 everyone_directory
drwxrwx--- 2 root  developers  4096 Nov 10 12:15 group_directory
-rw-rw---- 1 sammy sammy         15 Nov 10 17:07 group_modifiable
drwx------ 2 sammy sammy       4096 Nov 10 12:15 private_directory
-rw------- 1 sammy sammy        269 Nov 10 16:57 private_file
-rwxr-xr-x 1 sammy sammy      46357 Nov 10 17:07 public_executable
-rw-rw-rw- 1 sammy sammy       2697 Nov 10 17:06 public_file
drwxr-xr-x 2 sammy sammy       4096 Nov 10 16:49 publicly_accessible_directory
-rw-r--r-- 1 sammy sammy       7718 Nov 10 16:58 publicly_readable_file
drwx------ 2 root  root        4096 Nov 10 17:05 root_private_directory
```

Permissions “Triads”

Sample Pictures Properties

General | Sharing | Security | Location | Previous Versions

Object name: C:\Users\Public\Pictures\Sample Pictures

Group or user names:

- Administrators (win7\Administrators)
- INTERACTIVE
- SERVICE

To change permissions, click Edit. Edit...

Permissions for CREATOR OWNER

	Allow	Deny
Full control		
Modify		
Read & execute		
List folder contents		
Read		
Write		

For special permissions or advanced settings, click Advanced. Advanced

Learn about access control and permissions

OK | Cancel | Apply

Permissions for Sample Pictures

Security

Object name: C:\Users\Public\Pictures\Sample Pictures

Group or user names:

- CREATOR OWNER
- SYSTEM
- BATCH
- Administrators (win7\Administrators)
- INTERACTIVE
- SERVICE

Add... | Remove

Permissions for CREATOR OWNER

	Allow	Deny
Full control		
Modify		
Read & execute		
List folder contents		
Read		

Learn about access control and permissions

OK | Cancel | Apply

General model of DAC systems

- We need to model the following entities:
  - **Subjects** who can exercise privileges (a.k.a., rights)
  - **Objects** on which privileges are exercised
  - **Actions** which can be exercised
- **Protection state**: a triple (S, O, A)
  - A: matrix with S rows and O columns
  - $A[s,o]$: privileges of subject s over object o

	file1	**file2**	**directoryX**
Alice	Read	Read, Write, Own	
Bob	Read, Write, Own	Read	Read, Write, Own
Charlie	Read, Write		Read

Lampson ‘71, Graham and Denning ‘72, **H**arrison, **R**uzzo and **U**llmann ‘76 -> **HRU model**

Transitions in the HRU model

- **Basic operations**
  - create (or destroy) subject <*s*>
  - create (or destroy) object <*o*>
  - add (or remove) <*permission*> into *[s,o]* matrix
- **Transitions**: sequences of basic operations
  - *"create file (subject u; file f)"*:
    - create object *f*
    - add “own” into *[u,f]*
    - add “read” into *[u,f]*
  - Is this right?

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  - *"create file (subject u; file f)"*:
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    - add “own” into *[u,f]*
    - add “read” into *[u,f]*
  - Is this right? No, we need to check if *f* existed before, otherwise *u* would be stealing it away!
  - We need an “if” construct for modeling transitions

Safety problems

- From an initial configuration, given a sequence of transitions, can s obtain a right r on f?
  - Obviously, yes if the owner o allows it!
  - But, if the owner does not?
    - If it happens, set of commands **unsafe by design**!
- More formally
  - Given an initial protection state and set of transitions, is there **any sequence of transitions that leaks** a certain right r (for which the owner is removed) into the access matrix?
  - If not, then the system is safe with respect to right r
- In a generic HRU model (with infinite resources): **undecidable** problem
  - Decidable in mono-operational systems, (substantially useless, e.g., you cannot create a file and own it)
  - .. or <u>if subjects/objects are finite</u>.

Common DAC Implementations

- Reproduction of HRU models
- Access matrix is a sparse matrix
- Alternative implementations:
  - **Authorizations table:** records non-null triples S-O-A, typically used in DBMS
  - **Access Control Lists:** records by column (i.e., for each **object**, the list of subjects and authorizations)
  - **Capability Lists:** records by row (i.e., for each **subject**, the list of objects and authorizations)

Access Control vs Capability Lists

Access Control Lists

- Focus on the object
- ACLs ≡ columns of the access control matrix

fun.com	Alice: {exec}	Bill: {exec,read,write}

Capability Lists

- Focus on the subject
- Capabilities ≡ rows of the access control

Alice	edit.exe: {exec}	fun.com: {exec,read}

ACLs vs Capability Lists

ACLs

- efficient with **per object** operations
- Most common case
- Some systems (e.g., POSIX) use abbreviated ACLs
- Cannot have multiple owners (partially achievable via groups).

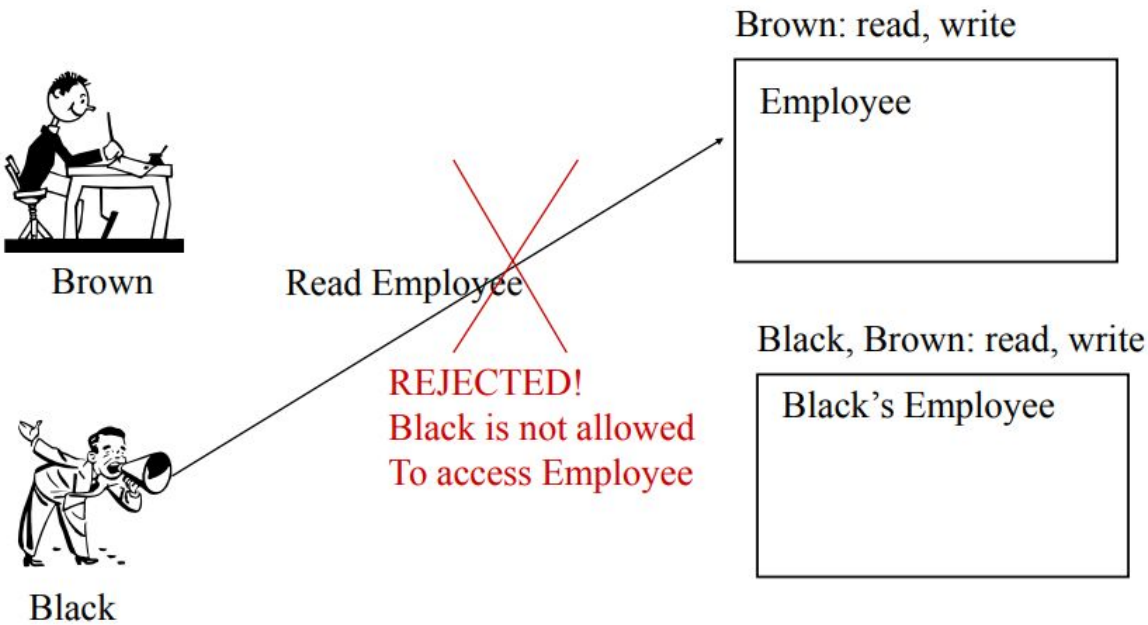
Capabilities

- efficient with **per subject** operations
- Usually **objects** change and **subjects** stay, so inefficient
- Capabilities are optional in POSIX (Linux and BSD).

General DAC shortcomings

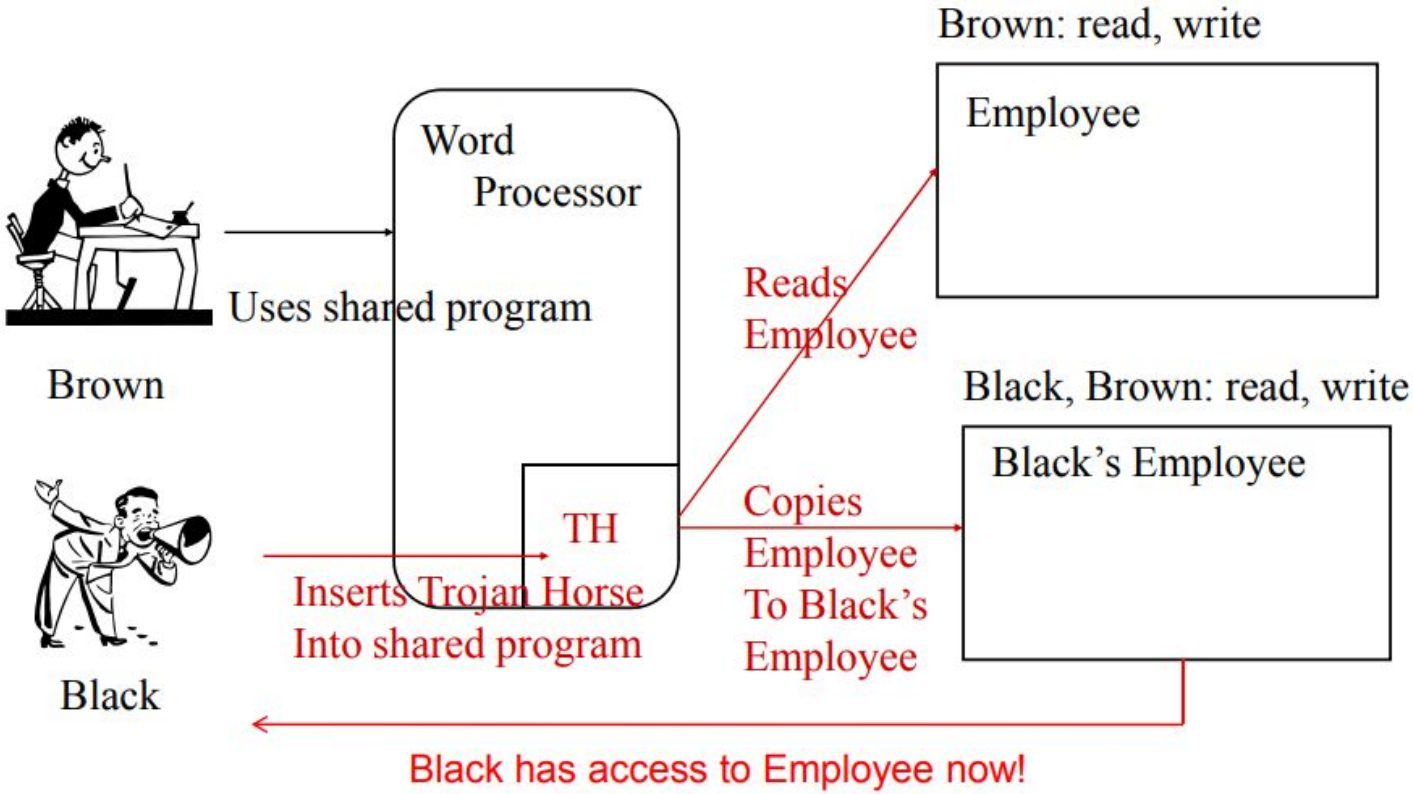
- Cannot prove safety
- Control access to objects but not to the data inside objects (granularity)
  - Susceptible to *malicious user* problem
  - Susceptible to *trojan horse* problem: malicious program running with privileges of the user
- Problems of <u>scalability and management</u>
  - each user-owner can potentially compromise security of the system with their own decisions

DAC and Trojan Horse



credits: https://www.comp.nus.edu.sg/~tankl/cs5322/slides/mac2.pdf

DAC Trojan Horse Problem



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Mandatory Access Control (MAC)

Idea: do not let owners assign privileges.

Privileges are set by a security **administrator**:

- E.g., defines a **classification** of *subjects* (or "clearance") and *objects* (or "sensitivity").

The **classification** is composed of:

- A strictly ordered set of **secrecy levels.**
- A set of **labels.**

SELinux Administration

File Help

Select:

- Status
- Boolean
- File Labeling
- User Mapping
- SELinux User
- Translation
- Network Port
- Policy Module

Revert Customized

(example of MAC rules in SELinux)

Filter

Active	Module	Description	Name
☐	apache	Allow httpd to act as a FTP server by listening on the	httpd_enable_ftp_server
☐	apache	Allow HTTPD to run SSI executables in the same dom	httpd_ssi_exec
☐	apache	Allow Apache to communicate with avahi service via (	allow_httpd_dbus_avahi
☑	apache	Allow httpd to use built in scripting (usually php)	httpd_builtin_scripting
☐	apache	Allow http daemon to send mail	httpd_can_sendmail
☐	apache	Allow httpd to access nfs file systems	httpd_use_nfs
☑	apache	Unify HTTPD to communicate with the terminal. Nee	httpd_tty_comm
☐	apache	Allow Apache to use mod_auth_pam	allow_httpd_mod_auth_ntlm_winbind
☐	apache	Allow HTTPD scripts and modules to connect to the r	httpd_can_network_connect
☑	apache	Unify HTTPD handling of all content files	httpd_unified
☐	apache	Allow apache scripts to write to public content. Dire	allow_httpd_sys_script_anon_write
☑	apache	Allow httpd to read home directories	httpd_enable_homedirs
☐	apache	Allow Apache to modify public files used for public fil	allow_httpd_anon_write
☐	apache	Allow Apache to use mod_auth_pam	allow_httpd_mod_auth_pam
☐	apache	Allow httpd to access cifs file systems	httpd_use_cifs
☑	apache	Allow httpd cgi support	httpd_enable_cgi
☐	apache	Allow HTTPD scripts and modules to network connect	httpd_can_network_connect_db
☐	apache	Allow httpd to act as a relay	httpd_can_network_relay
☐	bind	Allow BIND to write the master zone files. Generally t	named_write_master_zones
☐	cdrecord	Allow cdrecord to read various content. nfs, samba, r	cdrecord_read_content
☐	cron	Enable extra rules in the cron domain to support fcro	fcron_crond
☐	cvs	Allow cvs daemon to read shadow	allow_cvs_read_shadow
☑	domain	Allow unlabeled packets to work on system	allow_unlabeled_packets
☐	exim	Allow exim to connect to databases (postgres, mysq	exim_can_connect_db
☐	exim	Allow exim to create, read, write, and delete unprivile	exim_manage_user_files
☐	exim	Allow exim to read unprivileged user files.	exim_read_user_files
☐	ftp	Allow ftp to read and write files in the user home dire	ftp_home_dir
☐	ftp	Allow ftp servers to login to local users and read/writ	allow_ftpd_full_access
☐	ftp	Allow ftp servers to use nfs used for public file trans	allow_ftpd_use_nfs

Secrecy Levels (US example)

Top Secret

>

Secret

>

For Official Use Only (FOUO)

>

Unclassified

Secrecy Levels (NATO example)

COSMIC Top Secret

>

NATO Secret

>

NATO Confidential

>

Unclassified

Example (labels)

Policy

Energy

Finance

ATOMAL

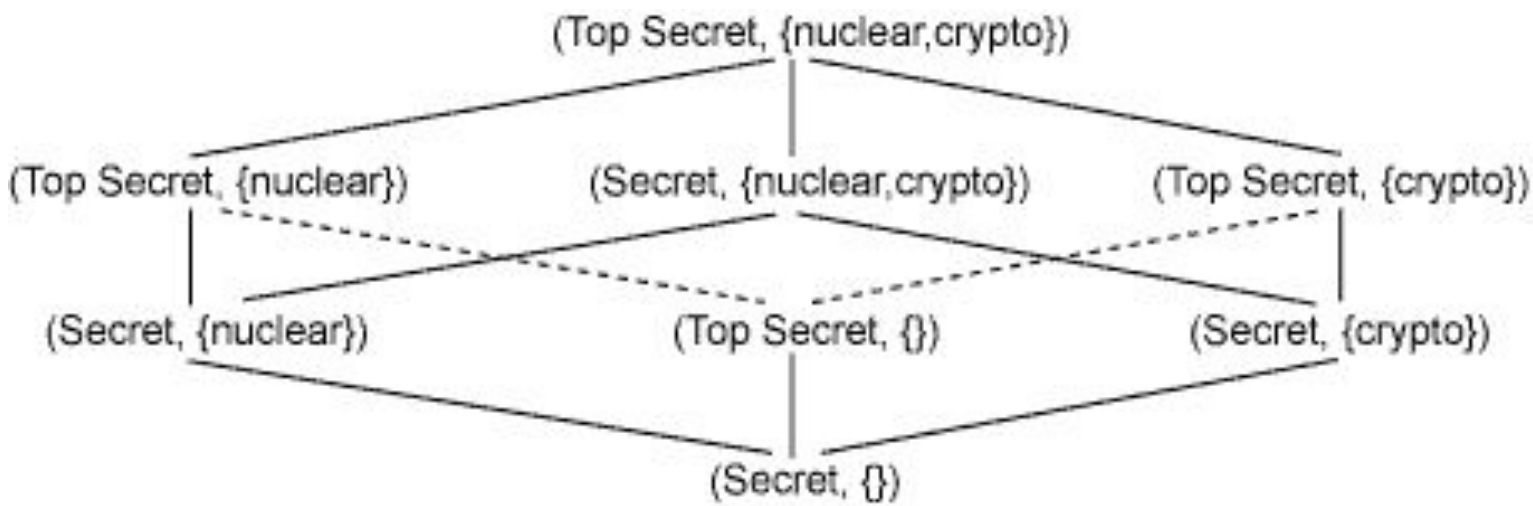
NOFORN

Secrecy Levels + Labels = Lattice (LBAC)

Classification = *partial order relationship*.

Dominance in a lattice is defined as:

$\{C1,L1\} \geq \{C2,L2\} \Leftrightarrow C1 \geq C2$ and $L2 \subseteq L1$



(reflexive, transitive, antisymmetric property)

Bell-LaPadula Model (BLP) - 1

Defines two MAC rules:

1. **Rule 1** (no read up, "simple security property")
   A subject s at a given secrecy level **cannot read** an object o at a **higher** secrecy level.
1. **Rule 2** (no write down, "star property")
   A subject s at a given secrecy level **cannot write** an object o at a **lower** secrecy level.

Defines one DAC rule:

Rule 3 (Discretionary Security Property) states the use of an access matrix to specify the discretionary access control.

Bell-LaPadula Model (BLP) - 2

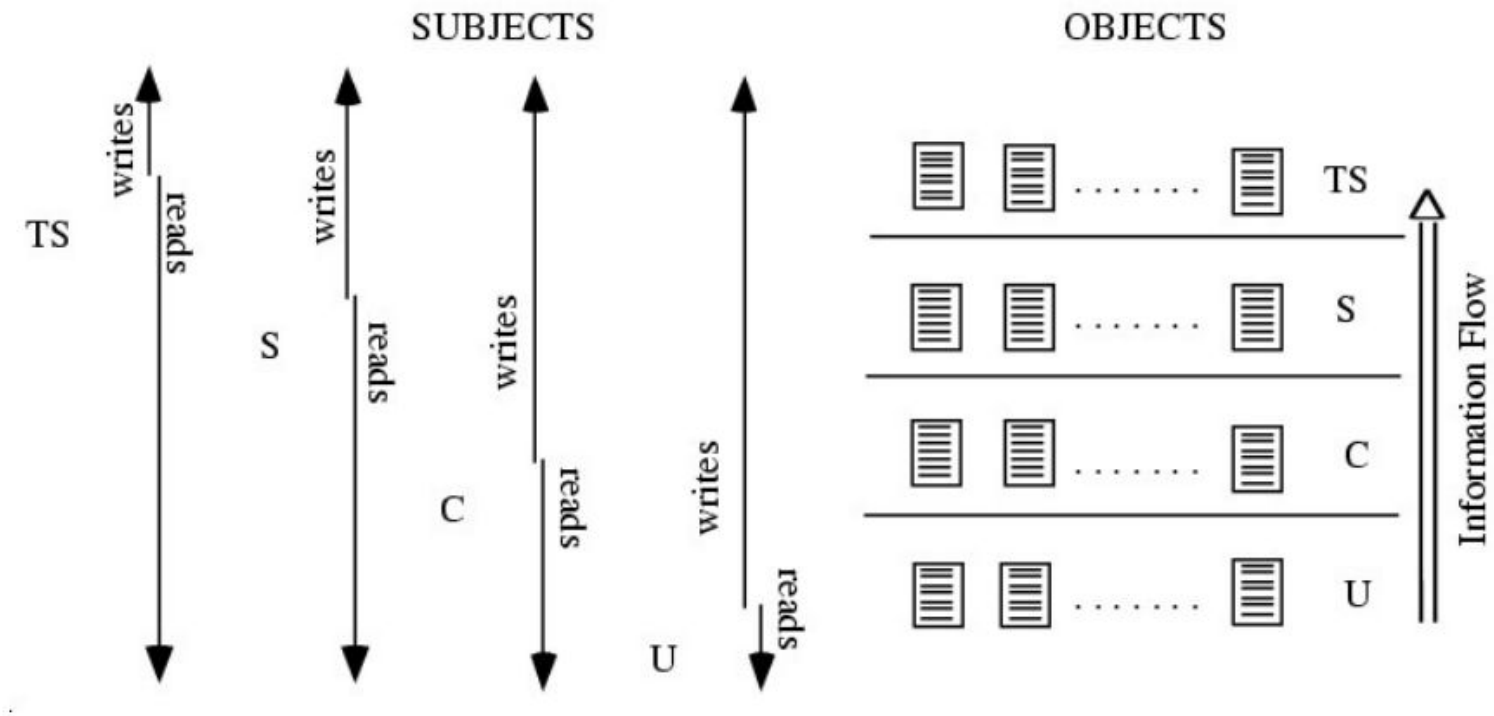
Tranquility property: secrecy levels of objects cannot change dynamically.

Result: monotonic flow of information toward higher secrecy levels

- need of **trusted subjects** who can declassify or sanitize documents

Limitations: does not address integrity. There are other models for integrity, e.g. http://en.wikipedia.org/wiki/Biba_Model

MAC Information Flow



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Conclusions

Access control, or authorization, defines *subjects*, *objects*, and *actions* in a system.

Access control **models** define how actions are (un)assigned to subjects and objects.

DAC are more common and “natural” than **MAC**, but can coexist.